Koorosh Roohi

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Summary

ML researcher and engineer; Ph.D. candidate at the University of Toronto and Vector Institute affiliate, working on efficient high-resolution inference for diffusion models and on multi-modal sensing. Previously led AI at Masterly (VLM/LLM agents).

Experience

Vector Institute Toronto, Canada
Faculty Affiliate Researcher, Generative AI *Mar 2026 – Present*

- Research on optimizing diffusion-model inference for high-resolution output: training-free methods that let pretrained text-to-image models generate beyond their native resolution while keeping attention cost and structural artifacts in check.
- Work hands-on with text-to-image models (FLUX, Qwen-Image, Ideogram), their Diffusers implementations and weights.

Masterly Inc. Toronto, Canada
AI Manager *Nov 2023 – Sep 2026*

- Led a team of **5** developers, taking **2** AI modules from design to production on AWS (SageMaker, Bedrock, Lambda, EC2).
- Designed Masterly Earendel, an agentic VLM video-analysis system that detects critical events, issues real-time alerts and refines its own context from past results.
- Built the company's MCP host and MCP servers, giving its LLM/VLM agents access to internal tools and data.
- Led development of Masterly FinSuite, an LLM-driven financial portal for reporting and decision support.

University of Toronto Toronto, Canada
Graduate Research Assistant *Apr 2023 – Present*

- Developing deep learning models that estimate infection exposure risk from indoor localization and activity recognition.
- Proposed a dynamic multi-modal fusion architecture for activity recognition from mmWave radar point clouds, reaching about **10%** higher accuracy than single-modality models.
- Built multi-modal datasets (mmWave radar, RGB-D, IMU, BLE) for activity recognition and indoor localization.

Amirkabir University of Technology Tehran, Iran
Research Assistant *Dec 2021 – Mar 2023*

- Built a gait-based person re-identification system (CNN with Horizontal Pyramid Matching, triplet loss) reaching **86%** accuracy on OUMVLP-Pose; deployed YOLOv5 with DeepSORT for multi-object tracking.

Education

University of Toronto Toronto, Canada
Ph.D. in Biomedical Engineering, GPA 4.0/4.0 *Sep 2023 – Aug 2027 (expected)*

- Thesis: Infection Risk Estimation via Multi-modal Sensing and Deep Learning in Healthcare

Amirkabir University of Technology (Tehran Polytechnic) Tehran, Iran
B.Sc. in Computer Engineering, GPA 3.85/4.0 *Sep 2018 – Mar 2023*

Selected Publications

- **WACV 2027** PhotoQuilt: Training-Free Arbitrary-Resolution Photomosaics via Bootstrapped Tiled Denoising
- **NeurIPS 2026** SEGA: Spectral-Energy Guided Attention for Resolution Extrapolation in Diffusion Transformers
- **arXiv 2026** AReA: Adaptive Receptive-Field Attention for Efficient High-Resolution Diffusion Transformers
- **EMBC 2025** Using mmWave Radar and Deep Learning to Classify Caregiver Activities for Infection Prevention
- **Signal, Image and Video Processing 2025** FUVT: A Deep Few-shot Unsupervised Learning-based Video-to-Video Translation Scheme using Kalman Filtering and Relativistic GAN

Technical Skills

- **Machine Learning**: PyTorch, Hugging Face Diffusers, TensorFlow, scikit-learn, DiT, VLMs, LLM agents, MCP
- **Sensing & Vision**: mmWave radar point clouds, sensor fusion (RGB-D, IMU, BLE), OpenCV, YOLO, DeepSORT
- **Engineering**: Python, C#/.NET, SQL, NumPy, Pandas, AWS (SageMaker, Bedrock, Lambda, EC2), Docker, Git, Linux

Awards & Service

- **Awards**: Graduate Scholarships in Science and Technology, Mount Sinai ($15K, 2026); EPIC Doctoral Award ($10K, 2025)
- **Reviewer**: ICML 2026, NeurIPS 2026, WACV 2027, ICLR 2027; *Circuits, Systems, and Signal Processing* (13 reviews)